coursera | Northeastern University

Foundations of Statistical Learning & Algorithms

Today's goal • 3/3 items

Module 1
Introduction to Linear Algebra and Vector Spaces

Module 2
Linear Transformation and Matrix Decomposition

Linear Independence
- Understanding Linear Independence in Vector Spaces — Reading • 18 min
- Linear Independence, Basis, and Rank — Ungraded App Item • 20 min
- Linear Independence — Reading • 30 min

Linear Mapping
- Exploring Transformations: Understanding Linear Mappings — Reading • 1h
- Linear Mapping: Part 1 — Video • 8 min
- Linear Mapping: Part 2 — Video • 8 min
- Linear Mapping — Reading • 5 min
- Matrix Representation — Reading • 6 min
- Practice Quiz: Linear Mapping — Graded Assignment • 18 min

Lesson 3: Eigenvalues and Eigenvectors
- Introduction to Eigenvalues and Eigenvectors — Reading • 40 min
- Eigenvectors and Eigenvalues: Examples and Applications — Reading • 30 min

Lesson 4: Cholesky Decomposition
- Cholesky Decomposition — Reading • 5 min
- Solved Examples — Reading • 10 min

Lesson 5: Singular Value Decomposition
- Introduction to Singular Value Decomposition (SVD) — Reading • 28 min
- Derivation of Singular Value Decomposition (SVD) from Eigenvalues and Eigenvectors — Reading • 15 min

Module 3
Probability Foundations for Statistical Learning

Module 4
Introduction to Model Evaluation and Optimization

Linear Independence

0:04 / 1:24 2x

Linear Combination

Given a set of vectors $X_1, X_2, \ldots X_k$ in a vector space V and a set of scalers $\lambda_1, \lambda_2, \ldots..\lambda_k$ from the real numbers $\mathbb{R}$, a vector is called a linear combination of the vectors $X_1, X_2, \ldots X_k$ if it can be expressed as:

$$\mathcal{V} = \lambda_1 x_1 + \lambda_2 x_2 + \ldots\ldots\ldots\ldots \lambda_k x_k \text{ or}$$

$$v = \sum_{i=1}^{k} \lambda_i x_i$$

This interactive Linear Combination Visualizer GeoGebra activity provides a visual and interactive exploration of linear independence and dependence in vector spaces. It allows you to manipulate vectors and observe how their linear combinations affect their independence, helping to deepen your understanding of these fundamental concepts in linear algebra.

Linear Dependence

Linear dependence occurs when a set of vectors can be expressed as a linear combination of each other. If there exists a non-trivial combination of vectors $x_1, x_2, \ldots x_k$ such that $\mathbf{0} = \sum_{i=1}^{k} \lambda_i \boldsymbol{x}_i$ with at least one $\lambda_i \neq 0$, the vectors are linearly dependent.

Additional Recommended Resource

- Statlect: Linear Dependence: Comprehensive exploration of linear dependence, discussing its implications in statistics and vector spaces.

Go to next item ✓ Completed

Like Dislike Report an issue

Go to next item →